



BY 508: Practical Metabolomics

Course Code	BY 508
Course Name	Practical Metabolomics
L-T-P-C	1-0-2-3
Prerequisites	IC136, or consent of faculty; MS, MSc, PhD from Basic Sciences
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Module 1**

Theory and Tutorials: Metabolites and metabolite profiling, Metabolomics - applications and its role in systems biology with case studies, Targeted and untargeted metabolomics, General work flow including quenching and sample preparation, Detection and quantification of metabolites by advanced analytical techniques (NMR/Mass spectroscopy, HPLC). Statistical methods (PCA, PLS, PLS-DA) in metabolomics. Pathway and metabolome databases. Software tools available for metabolomics analysis [10 Lectures]

- **Module 2**

Practicals – mini-project involving metabolite profiling and metabolomics experiments based on Module 1 contents. Review and development of Protocols/workflow, Design and conduct of experiments, analytical techniques (NMR, MS, HPLC), Data handling using relevant software, Statistical analysis. Scientific report writing in metabolomics for evaluation. [32 Lectures]

References

1. J.Nielsen and M.C. Jewett (Editors), **Metabolomics – A powerful Tool in Systems Biology**, Springer.
2. Dr. Silas G. Villas et al., **Metabolome Analyses: an Introduction**, John Wiley & Sons.
3. Additional reference material and scientific papers will be provided.